

Ensemble Methods

TD – PAC-Bayesian Learning

M2 Computer Science – MALIA

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Abstract. This tutorial accompanies Sections 2–7 of the lecture notes *PAC-Bayesian Learning: a Theoretical Framework for Ensemble Methods*. The exercises are of increasing difficulty; those marked $(\star)$ are at Master level. Notation: $S \sim \mathcal{D}^m$, voters $h : \mathcal{X} \rightarrow \{-1, +1\}$, posterior Q , prior P , Gibbs classifier G_Q , majority vote B_Q , $W_Q(x, y) = \mathbb{E}_{h \sim Q} \mathbb{1}[h(x) \neq y]$, joint error $e_Q = \mathbb{E} W_Q^2$, disagreement d_Q .

Exercise 1 – The Kullback–Leibler divergence

Let $\mathcal{H} = \{h_1, \dots, h_n\}$ be finite and P, Q two distributions on $\mathcal{H}$ with $P_i > 0$.

1. Using Jensen's inequality, show that $\text{KL}(Q \| P) \geq 0$, with equality iff $Q = P$.
2. Assume P uniform. Show that $\text{KL}(Q \| P) = \ln n - H(Q)$ where $H(Q) = -\sum_i Q_i \ln Q_i$. What are the minimal and maximal values of $\text{KL}(Q \| P)$? Compute it when Q is uniform over $k \leq n$ voters.
3. Show that $\text{KL}(Q \otimes Q \| P \otimes P) = 2 \text{KL}(Q \| P)$.
4. Show that $\text{KL}(\mathcal{N}(\mu_1, \sigma^2 I_d) \| \mathcal{N}(\mu_0, \sigma^2 I_d)) = \frac{\|\mu_1 - \mu_0\|^2}{2\sigma^2}$.
5. $(\star)$ Binary kl. Fix $q \in [0, 1]$. Show that $p \mapsto \text{kl}(q \| p)$ is convex on $(0, 1)$, minimal at $p = q$, and prove Pinsker's inequality $\text{kl}(q \| p) \geq 2(q - p)^2$ by studying $g(p) = \text{kl}(q \| p) - 2(q - p)^2$.

Exercise 2 – Change of measure and Gibbs posterior

Let $\phi : \mathcal{H} \rightarrow \mathbb{R}$ and P a distribution on a finite set $\mathcal{H}$. Define $Q_\phi(h) = \frac{P(h)e^{\phi(h)}}{\mathbb{E}_P e^\phi}$.

1. Show that for any Q , $\text{KL}(Q \| Q_\phi) = \text{KL}(Q \| P) - \mathbb{E}_Q \phi + \ln \mathbb{E}_P e^\phi$.
2. Deduce the change of measure inequality $\mathbb{E}_Q \phi \leq \text{KL}(Q \| P) + \ln \mathbb{E}_P e^\phi$ and the variational formula $\ln \mathbb{E}_P e^\phi = \max_Q \{\mathbb{E}_Q \phi - \text{KL}(Q \| P)\}$.
3. Let $\lambda > 0$. Show that the minimizer of $Q \mapsto \widehat{R}_S(G_Q) + \frac{1}{\lambda m} \text{KL}(Q \| P)$ is the Gibbs posterior $Q_\lambda(h) \propto P(h)e^{-\lambda m \widehat{R}_S(h)}$, and give the minimal value.
4. Describe Q_λ when $\lambda \rightarrow 0$ and $\lambda \rightarrow +\infty$. Interpret λ .
5. Recover Q_λ directly with a Lagrange multiplier for the constraint $\sum_h Q(h) = 1$.

Exercise 3 – From the general theorem to classical bounds

Recall the general theorem: for convex Δ , w.p. $\geq 1 - \delta$, for all Q , $\Delta(\hat{R}_S(G_Q), R_{\mathcal{D}}(G_Q)) \leq \frac{1}{m} [\text{KL}(Q\|P) + \ln \frac{1}{\delta} \mathbb{E}_{S'} \mathbb{E}_{h \sim P} e^{m\Delta(\hat{R}_{S'}(h), R_{\mathcal{D}}(h))}]$.

1. Explain precisely where each of the following is used in the proof: convexity of Δ , the change of measure, Markov's inequality, the independence of P from S .
2. (*Hoeffding-type bound*). Take $\Delta(q, p) = \lambda(p - q)$ for a fixed $\lambda > 0$. Using Hoeffding's lemma, show that w.p. $\geq 1 - \delta$, for all Q : $R_{\mathcal{D}}(G_Q) \leq \hat{R}_S(G_Q) + \frac{\text{KL}(Q\|P) + \ln \frac{1}{\delta}}{\lambda m} + \frac{\lambda}{8}$.
3. Minimize the right-hand side in λ . Why is it not allowed to plug this value into the bound? Propose a correct way to optimize λ and give the resulting price.
4. (*Catoni's bound*). Let $C > 0$ and $\mathcal{F}_C(p) = -\ln(1 - p(1 - e^{-C}))$. Show that for a fixed voter with risk p and the 0-1 loss, $\mathbb{E}_{S'} e^{-Cm\hat{R}_{S'}(h)} = (1 - p(1 - e^{-C}))^m$. Deduce Catoni's bound.
5. Numerical application: $m = 1000$, $\delta = 0.05$, $\hat{R}_S(G_Q) = 0.1$, $\text{KL}(Q\|P) = 5$. Compute McAllester's bound $\hat{R}_S + \sqrt{(\text{KL} + \ln \frac{2\sqrt{m}}{\delta})/(2m)}$ and the Hoeffding-type bound for $\lambda \in \{0.1, 0.2\}$.

Exercise 4 – Occam's razor as a PAC-Bayesian bound

1. Let $\mathcal{H}$ be countable and $Q = \delta_h$ a Dirac on $h \in \mathcal{H}$. Compute $\text{KL}(\delta_h\|P)$. What does Seeger's bound give for a single voter? Compare with the Occam bound of the lecture notes.
2. Why can we not use Dirac posteriors when $\mathcal{H}$ is uncountable (e.g. linear classifiers with a Gaussian prior)?
3. Let $\mathcal{H} = \{h_1, h_2, \dots\}$ be countable and $P(h_k) = \frac{6}{\pi^2 k^2}$. Check that P is a probability and give the complexity term of h_k . Interpret this prior in terms of *description length*.

Exercise 5 – Bounds on the risk of the majority vote

We assume voters in $\{-1, +1\}$, 0-1 loss, and count ties as errors.

1. Show that $R_{\mathcal{D}}(B_Q) = \mathbb{P}(W_Q \geq \frac{1}{2})$ and $R_{\mathcal{D}}(G_Q) = \mathbb{E}W_Q$. Deduce $R_{\mathcal{D}}(B_Q) \leq 2R_{\mathcal{D}}(G_Q)$. Give an example of distribution of W_Q for which this inequality is an equality.
2. Show that $e_Q = \mathbb{E}[W_Q^2]$, $d_Q = \mathbb{E}[2W_Q(1 - W_Q)]$, and $R_{\mathcal{D}}(G_Q) = e_Q + \frac{1}{2}d_Q$. Show also that $d_Q \leq 2R_{\mathcal{D}}(G_Q)(1 - R_{\mathcal{D}}(G_Q))$.
3. Prove the tandem bound $R_{\mathcal{D}}(B_Q) \leq 4e_Q$. Show that it is better than the first-order bound iff $d_Q \geq R_{\mathcal{D}}(G_Q)$.
4. For $\mu < \frac{1}{2}$, show that $R_{\mathcal{D}}(B_Q) \leq \frac{\mathbb{E}(W_Q - \mu)^2}{(1/2 - \mu)^2}$. Find the optimal μ^* and show that the optimal value is $\frac{\text{Var } W_Q}{\text{Var } W_Q + (1/2 - \mathbb{E}W_Q)^2}$ (Cantelli's inequality).
5. Deduce the C-bound $R_{\mathcal{D}}(B_Q) \leq 1 - \frac{(1 - 2R_{\mathcal{D}}(G_Q))^2}{1 - 2d_Q}$ (when $R_{\mathcal{D}}(G_Q) < \frac{1}{2}$).
6. (*Condorcet*). Q is uniform over n voters erring independently with probability p . Compute $R_{\mathcal{D}}(G_Q)$, e_Q , d_Q as functions of n and p . For $p = 0.3$, compute the exact risk of the vote and the three bounds for $n = 3$, then the C-bound for $n = 25$ and its limit when $n \rightarrow \infty$. Comment.

Exercise 6 – Linear classifiers with a Gaussian posterior

Voters $h_{\mathbf{v}}(x) = \text{sign}\langle \mathbf{v}, x \rangle$, prior $P = \mathcal{N}(\mathbf{0}, I_d)$, posterior $Q_{\mathbf{w}} = \mathcal{N}(\mathbf{w}, I_d)$.

1. Show that $B_Q(x) = \text{sign}\langle \mathbf{w}, x \rangle$.
2. Show that $\mathbb{P}_{\mathbf{v} \sim Q_{\mathbf{w}}}(h_{\mathbf{v}}(x) \neq y) = \Phi\left(-\frac{y\langle \mathbf{w}, x \rangle}{\|x\|}\right)$ where Φ is the standard Gaussian CDF. Give $\hat{R}_S(G_{Q_{\mathbf{w}}})$ and $\text{KL}(Q_{\mathbf{w}}\|P)$.
3. Show that, for a fixed C , minimizing Catoni's bound over $\mathbf{w}$ amounts to minimizing $C \sum_i \Phi(-y_i \langle \mathbf{w}, x_i \rangle / \|x_i\|) + \frac{1}{2} \|\mathbf{w}\|^2$. Compare with the SVM.
4. For $\mathbf{w} = s\mathbf{u}$ with $\|\mathbf{u}\| = 1$, describe the evolution of $\hat{R}_S(G_{Q_{s\mathbf{u}}})$ and of the KL when s grows. Relate to the notion of margin.

Exercise 7 – (★) The PAC-Bayes- λ bound and its minimization

1. Prove the refined Pinsker inequality $\text{kl}(q\|p) \geq \frac{(p-q)^2}{2p}$ for $q < p$.
2. Starting from Seeger's bound, show that w.p. $\geq 1 - \delta$, for all Q and all $\lambda \in (0, 2)$ simultaneously, $R_{\mathcal{D}}(G_Q) \leq \frac{\hat{R}_S(G_Q)}{1-\lambda/2} + \frac{\text{KL}(Q\|P) + \ln \frac{2\sqrt{m}}{\delta}}{\lambda(1-\lambda/2)m}$. Why is no union bound over λ needed here, contrary to Exercise 3?
3. For a fixed Q , compute the optimal λ . For a fixed λ , compute the optimal Q on a finite set of voters. Write the alternating minimization algorithm.
4. Explain why the posterior returned by this algorithm on a random forest may give a *worse* majority vote than the uniform one.

Exercise 8 – (★) Out-of-bag samples

Consider bagging with bootstrap samples of size k drawn with replacement among the m training examples.

1. Compute the probability that a given example is out-of-bag for a given tree; give its limit when $m \rightarrow \infty$ for $k = m$ and for $k = m/2$. Same question for being out-of-bag for two given trees.
2. Explain why the OOB loss $\hat{L}_t$ of tree t is an unbiased estimate of $R_{\mathcal{D}}(h_t)$ given S_t , and why a uniform prior over the *indices* of the trees is a legitimate PAC-Bayesian prior although the trees depend on S .
3. Show that if $\mathbb{E}e^{N \text{kl}(\hat{p}_N\|p)} \leq 2\sqrt{N}$ for all N , then for $n_0 \leq N$, $\mathbb{E}e^{n_0 \text{kl}(\hat{p}_N\|p)} \leq 2\sqrt{n_0}$. Why is this useful?

Exercise 9 – (★) Self-bounding learning

Let $\mathcal{B}(Q, S, \delta)$ be a bound such that $\mathbb{P}_S(\forall Q, R_{\mathcal{D}}(Q) \leq \mathcal{B}(Q, S, \delta)) \geq 1 - \delta$.

1. Let Q_S be the posterior returned by any algorithm run on S . Show that $\mathbb{P}_S(R_{\mathcal{D}}(Q_S) \leq \mathcal{B}(Q_S, S, \delta)) \geq 1 - \delta$. Why does this fail for a bound that holds only for a *fixed* predictor (e.g. Hoeffding's inequality)?
2. You build $K = 5$ random forests with tree depths 1, 2, 4, 8 and unlimited, and compute for each of them the OOB tandem certificate. How must you modify the certificates to be allowed to keep the forest with the smallest one? What is the cost in the complexity term?
3. Among the following choices, which are free, which require a union bound, which are forbidden? (a) the temperature λ of a Gibbs posterior; (b) the parameter C of Catoni's bound; (c) the variance of a Gaussian prior chosen on a grid of 10 values; (d) the centre

of a Gaussian prior learned on the sample used in the bound; (e) the number of gradient iterations when minimizing the bound.

4. Let $p = \overline{\text{kl}}^{-1}(q, \varepsilon)$. Show that $\frac{\partial p}{\partial \varepsilon} = \frac{p(1-p)}{p-q}$ and $\frac{\partial p}{\partial q} = \frac{p(1-p)}{p-q} \ln \frac{p(1-q)}{q(1-p)}$. Deduce the gradient of the first-order certificate $2\overline{\text{kl}}^{-1}(\rho^\top \widehat{L}, \frac{\text{KL}(\rho||\pi)+c}{n})$ with respect to ρ .
5. Why is the test-set bound usually tighter than a self-bounding certificate for a random forest? Give one advantage of the self-bounding approach.